

LUMAFLOW · DESIGN DECISIONS REGISTER

LumaFlow design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/lumaflow		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-01	Amish Chadha	Register opened with the build plan; budget treated as a value-engineering target

LumaFlow design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Accept the design-for-construction changes (window clamped from below, studs in the lower cap, screwed LED head, sealed tube seats, 40 mm upper cap, threaded ports, welded sensor boss, drilled bracket plate with pipe clamps and saddles, enclosure lid, 25 mm under the fins)	Accept all; accept some and revise others	Accept all	Every component	LMF-DDR-003, P1 to P11
2	How the LEDs are stopped when the LED head is off the cap (R12 requires it; the concept had no mechanism)	(a) a loop wire in the LED head's plug that the controller watches; (b) a microswitch pressed by the sink base; (c) procedure and the lid interlock only	(a)	LED head plug and wiring (build plan section 3.12.1)	LMF-DDR-003, A1
3	What the quartz window bears on	(a) metal to glass on the lapped stainless ring, as modelled; (b) a 0.5 mm PTFE washer under the window, with the optics re-run	(a) for the first prototype; inspect the window edge after the TRL 4 pressure test	Window stack (build plan section 3.5)	LMF-DDR-003, A2
4	Partner that can supply real well-water UV transmittance data and later test the reactor	A well-water association, a university lab or a water charity	None yet (partners are picked per area later)	Not part of the TRL 3 build	LMF-DDR-001 O1, LMF-DDR-002 O1
5	Dose bar on the enclosure in the product renders	Keep it and add it to the requirements; remove it from the appearance model	Keep it	Lid light pipe (a five-segment bar instead of one light)	REVIEW 2026-09-26, item 1
6	Acorn nuts on the tie rods (appearance model)	Acorn nuts; plain nuts	Acorn nuts	Already used in the constructable design	REVIEW 2026-09-26, item 2
7	Solenoid connector and port collars as drawn in the renders	Accept as catalogue-part detail; simplify	Accept	None	REVIEW 2026-09-26, item 3
8	Adapter and power cord in the hero render	Show the cord entering the cabinet wall; show the adapter	Show the cord, keep the adapter in the model and BOM	None	REVIEW 2026-09-26, item 4
9	Product label and UV-C warning label	Add a labels line to the BOM; leave out	Add a UV-C warning label line at the next BOM revision	A label on the lower cap	REVIEW 2026-09-26, item 5

#	Decision needed	Options	Recommendation	Affects in the build	Source
10	A brief LED pulse during long idle periods to limit growth in the reactor	Add a firmware rule; leave out	None yet	Firmware only; not part of the TRL 3 build	LMF-PRC-001, open questions

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The flow sensor and the valve have a 3/8 in push-fit port at each end and a flat face for the saddle, and their port centres match the stem height	The stem adaptors plug straight into them and the saddles steady them	LMF-DDR-003, P6 and P9
2	Flow coefficients (Kv) of the flow sensor and the valve	R8 rests on assumed values of 0.35 and 0.25	LMF-CAL-001 [G1]; LMF-DDR-002 E6
3	The heat sink's fin gaps are 7.5 mm or more, with nine fins on 90 x 90 mm	The M4 socket heads of the LED head sit in the gaps	LMF-DDR-003, P3
4	The LED board's output bin at 275 nm and its mounting holes (three on a 38 mm circle)	The dose rests on 60 mW per LED; the board screws to the sink	LMF-CAL-001 [B1]; BOM line 4
5	The pipe clamp's band and rubber are 3 mm thick or less in total	The tie rods pass 2 mm outside the clamp	LMF-DDR-003, P2
6	The stem adaptors are 1/4 BSPP male with a 3/8 in stem, stainless, food-contact	The cap ports are tapped 1/4 BSPP	LMF-DDR-003, P6
7	The PTFE liner supplier's reflectance data at 275 nm, wetted	R2 rests on 0.95; plain PTFE gives 46.3 mJ/cm ²	LMF-CAL-001 [B8]
8	Seal materials are food-contact EPDM and the O-ring size (about 61 mm inside, 1.78 mm section) is stocked	Wetted parts (R11)	LMF-DDR-003, P1 and P4

■ Value engineering

Value-engineering target: USD 340 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 402 (USD 62 over the target). Main cost drivers and savings worth trying:

- The largest lines are the six UV-C LEDs (USD 54), the machined 316 lower cap (USD 48), the high-reflectance PTFE liner (USD 45), the tube with its welded sensor boss (USD 38) and the quartz window (USD 32).
- Making the design constructable added USD 64: the welded sensor boss (USD 12 more on the tube line), the retaining ring and seals (USD 12), the threaded ports and stem adaptors (USD 10 more on fittings), the drilled bracket plate and saddles (USD 10 more), the pipe clamps (USD 8), more machining on both caps (USD 9), more fixings (USD 4) and drilling of the heat sink (USD 1), less USD 2 on the sensor line.
- Savings worth trying: a plain PTFE liner instead of the high-reflectance grade (most of USD 45, but the laminar dose falls to 46.3 mJ/cm², inside the 20 % margin, so R2 would be at risk); a clamp-on saddle with a band clamp instead of the welded boss (about USD 10); one machine shop quote for both caps, the tube boss and the holder together; LED prices at quantity, which fall fastest of all the lines.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D9: design for 90 %/cm water with alarm and valve closure below 40 mJ/cm ² , R3 kept visibly not met; budget to \$225; axial LED head with a flat quartz window; 275 nm LEDs; Hall-effect flow sensor; normally closed valve; vertical mounting with upward flow; external certified 24 V adapter; first user a well-water household	Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."	LMF-DDR-001
2026-09-25	TRL 3 items E1 to E8: 316 lower cap and shielded acetal upper cap; 50 mm bore, high-reflectance liner and 1.2 L/min; wall dose sensor; hold the budget and re-price; 15 Hz per L/min flow sensor; R8 excludes the restrictor; upstream pressure limiter; thermal cut-back	Amish: "i accept all your recommendations, go with them across all repos."	LMF-DDR-002
2026-09-26	Budget top-up to \$340 for the \$338.00 concept BOM	Amish: "I am ok with the budget top ups"	LMF-DDR-002, O4
2026-09-30	Make the design physically buildable while drawing the build plan	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	LMF-DDR-003 (open for review, open decision 1)
2026-09-30	Outstanding decisions go in this register, not in the build plan	Amish: "don't log outstanding decisions in this build plan - that is not the place for it. that should be in a separate design document logged and named as such"	This register
2026-10-01	budget_usd is a value-engineering target, reported as over or under, not as a limit	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens. its ok to ensure wording reflects that the hypothesis budget was x - the real cost being accrued is y"	LMF-CAL-001 v0.4, LMF-REQ-001 v0.6